Amazon Order Profitability Prediction

Using Machine Learning Classification Models

Course	ISOM 835 — Predictive Analytics and Machine Learning
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Dataset	Amazonsale_dataset.xlsx — Amazon Sales, Western US (2011-2015)
Submission	Friday, December 12th, 11:59 PM

3,203	**16**	**3**	**91.7%**
Orders	Features	ML Models	Best Accuracy

1. Executive Summary

This report presents a machine learning solution for predicting the profitability of Amazon sales orders in the Western United States. Using **Amazonsale_dataset.xlsx** — 3,203 transactions from 2011 to 2015 — three classification models were built and evaluated.

The best-performing model — **Gradient Boosting** — achieved **91.7% accuracy** and **ROC-AUC of 0.923**. Product category is the most influential factor: Tables (42.2%), Bookcases (40.0%), and Chairs (36.2%) generate the highest loss rates, while Paper, Art, and Appliances maintain 0% loss.

These insights enable targeted interventions: repricing high-risk furniture, expanding profitable categories, and deploying the model as a real-time order screening tool.

2. Introduction & Business Context

Business Problem

In e-commerce, not every sale is profitable. Heavy discounts, shipping costs, and low-margin product categories can erode profits even on large revenue orders. This project predicts which orders are likely to generate losses before fulfillment resources are committed.

Research Questions

1. Can we accurately predict whether an Amazon order will be profitable or result in a loss?
2. Which product categories carry the highest financial risk?
3. Which features are the strongest predictors of profitability?
4. How do Logistic Regression, Random Forest, and Gradient Boosting compare?

Dataset: Amazonsale_dataset.xlsx

The dataset contains Amazon sales records for the Western US from 2011–2015, with 3,203 order transactions across 17 product categories, 11 states, and 170 cities.

Attribute	Value
File Name	Amazonsale_dataset.xlsx
Total Records	3,203 orders
Features	16 (original + engineered)
Date Range	January 2011 — December 2015

States	11 (Western US)
Categories	17 product categories
Target	Profitable (1) / Loss (0)
Class Balance	89.4% Profitable / 10.6% Loss
Missing Values	None (0%)

Table 1: Dataset Summary — Amazonsale_dataset.xlsx

3. Exploratory Data Analysis

3.1 Target Variable Distribution

The dataset has a class imbalance: 2,863 profitable (89.4%) vs 340 loss orders (10.6%). F1-score and ROC-AUC are used as primary metrics instead of accuracy alone.

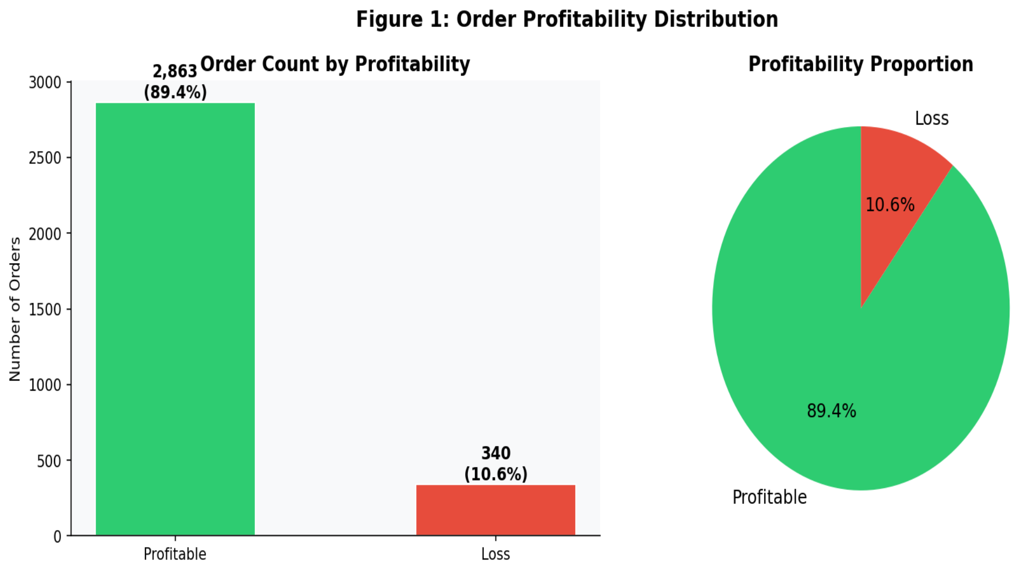

Figure 1: Order Profitability Distribution

3.2 Profitability by Category

Furniture dominates losses: Tables (42.2%), Bookcases (40.0%), Chairs (36.2%). Six categories — Paper, Art, Labels, Appliances, Copiers, Envelopes — have a perfect 0% loss rate.

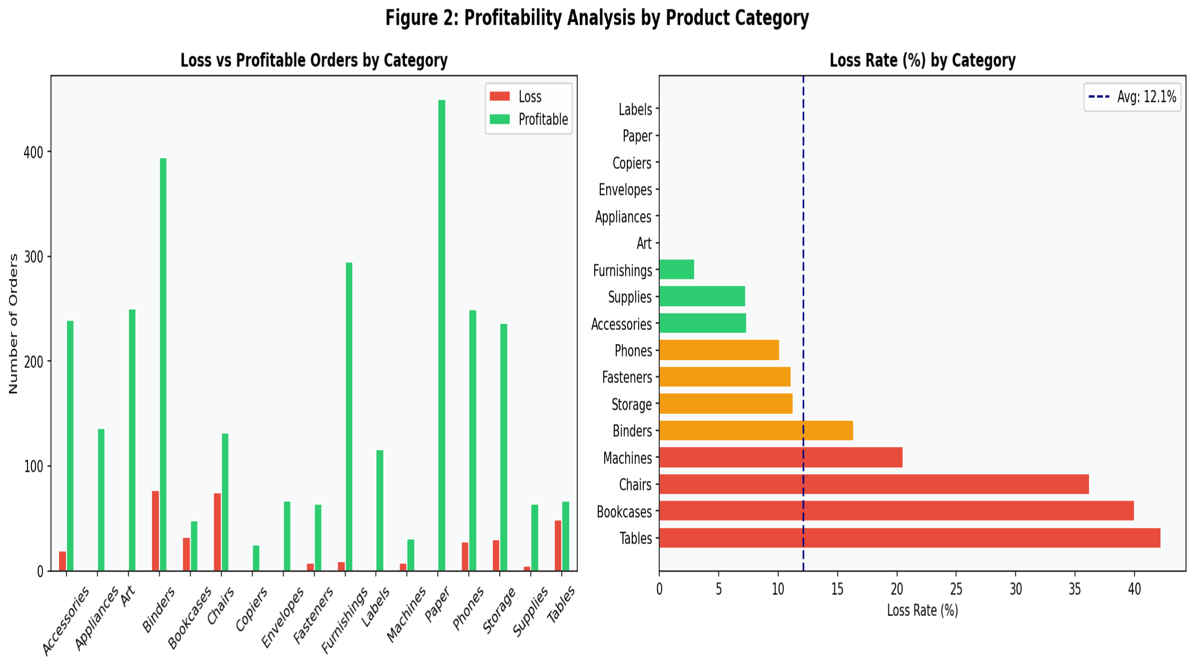

Figure 2: Profitable vs Loss Orders by Category

3.3 Sales & Profit Distributions

Sales are right-skewed with most orders under $500. High sales alone do not guarantee profitability — 340 orders fall below the break-even line.

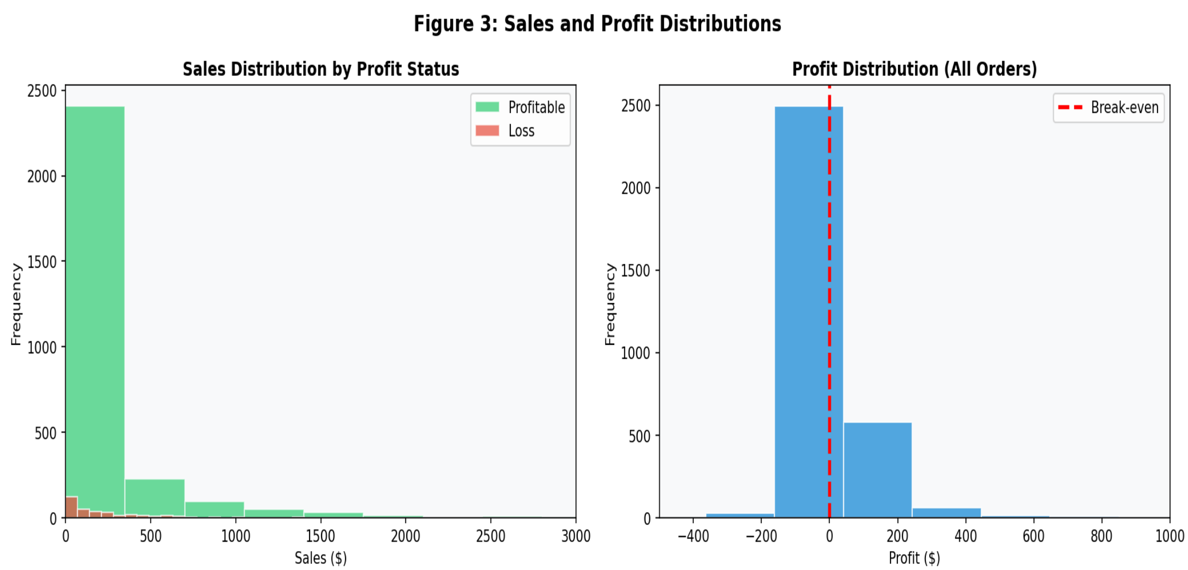

Figure 3: Sales and Profit Distributions

3.4 Feature Correlations

Profit and Profit_Margin strongly correlate with the target. Among model features, Sales shows the highest positive correlation; Shipping_Days and Month show weaker signals.

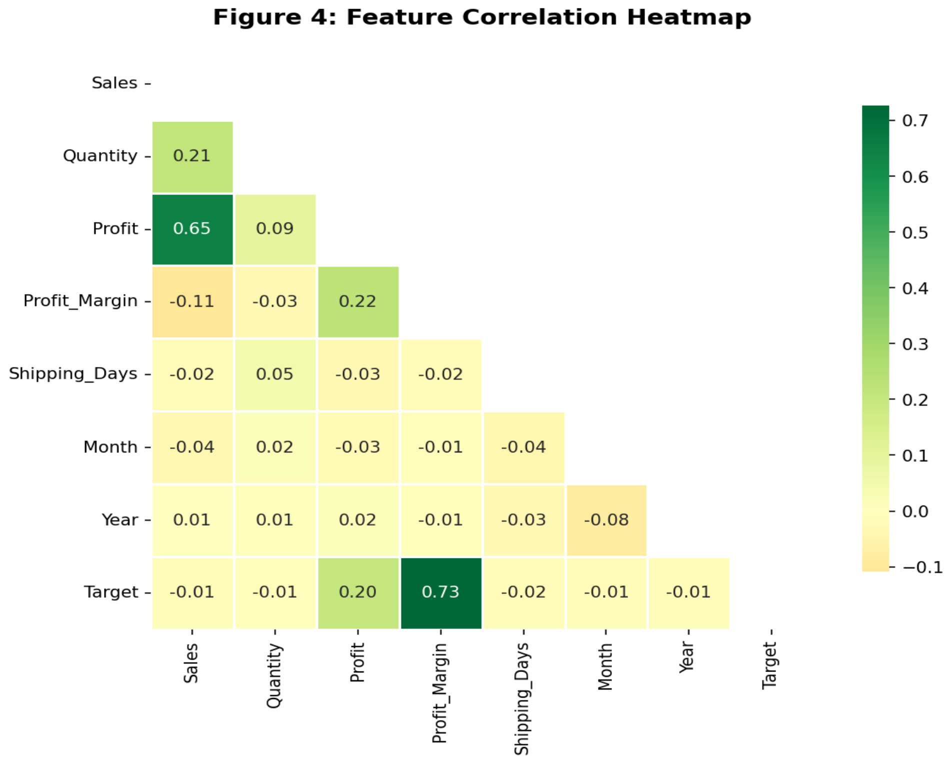

Figure 4: Feature Correlation Heatmap

3.5 Geographic Loss Patterns

Colorado shows the highest loss rate among states with enough data. California (62% of records) stays near the 10.6% average. Utah and Nevada have the lowest loss rates.

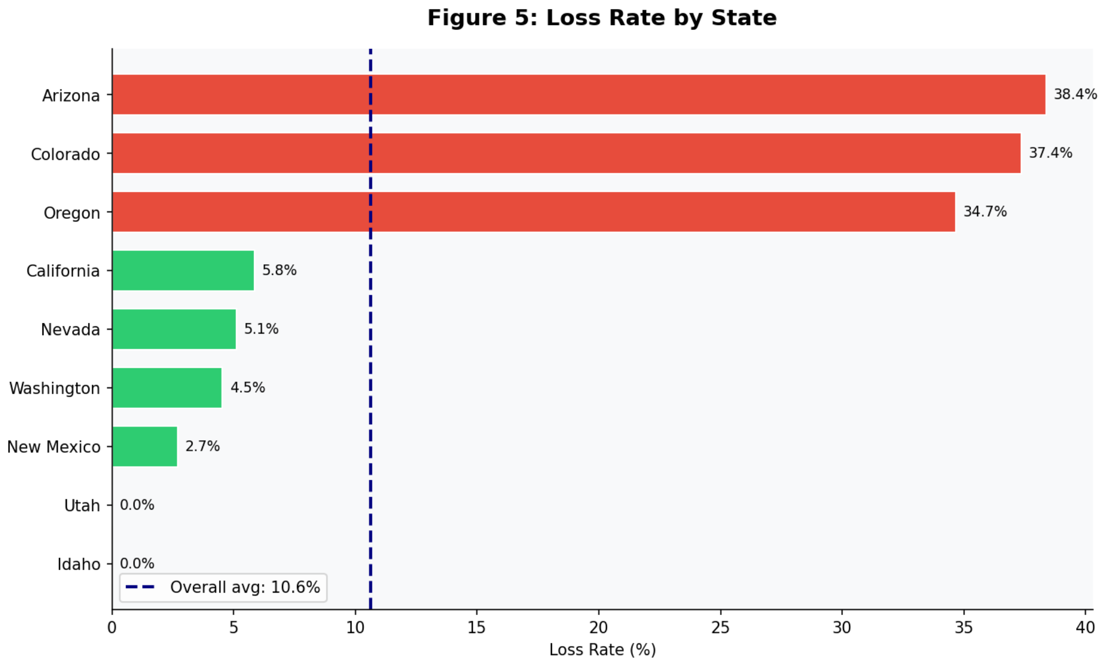

Figure 5: Loss Rate by State

3.6 Profit Margin by Category

Furniture categories have both high loss rates and high margin variance — indicating inconsistent discounting. Copiers and Machines have high median margins but occasional deep losses.

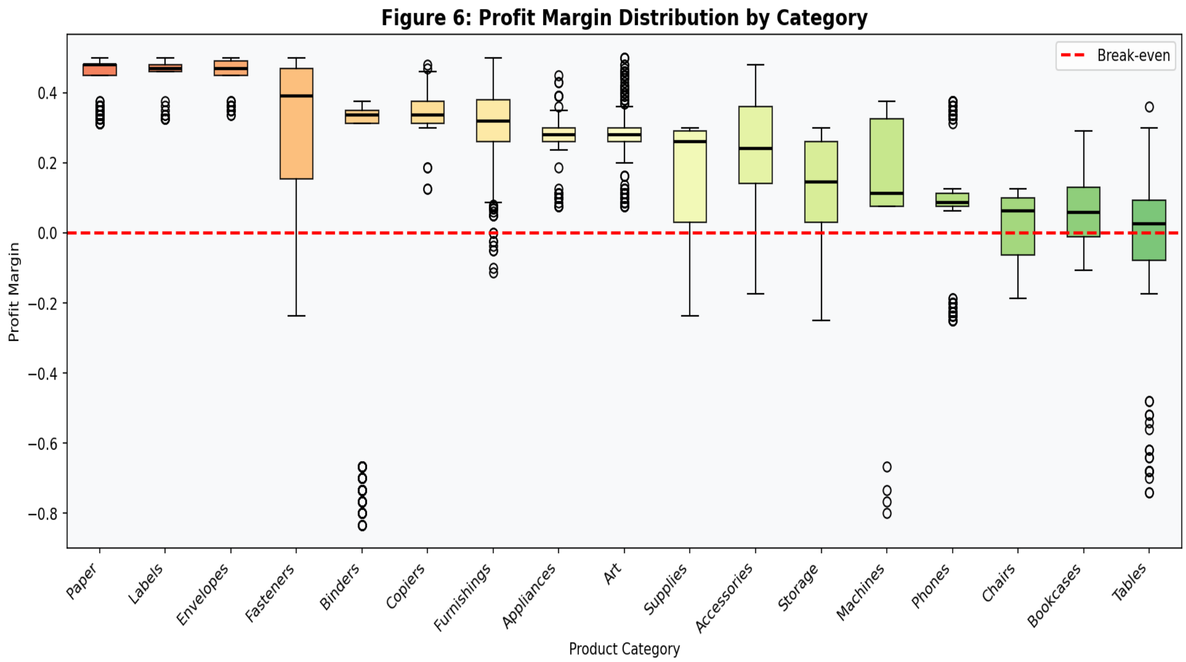

Figure 6: Profit Margin Distribution by Category

4. Methodology

4.1 Data Preprocessing

- **Target Encoding:** Profit_Status converted to binary (1=Profitable, 0=Loss).
- **Categorical Encoding:** Category, State, Sales_Size encoded with LabelEncoder.
- **Feature Scaling:** StandardScaler applied for Logistic Regression.
- **Train/Test Split:** 80/20 stratified split (2,562 train / 641 test).
- **No imputation needed:** Amazonsale_dataset.xlsx has zero missing values.

4.2 Features Used

Feature	Type	Rationale
Sales ($)	Numerical	Revenue drives cost coverage
Quantity	Numerical	Volume affects unit economics
Shipping Days	Numerical	Fulfillment cost proxy
Order Month	Numerical	Seasonal pricing effects
Order Year	Numerical	Market maturity over time
Product Category	Categorical (enc.)	Strongest profitability driver
Customer State	Categorical (enc.)	Regional logistics cost differences
Sales Size Tier	Categorical (enc.)	Low/Medium/High order classification

Table 2: Feature Engineering Summary

4.3 Models & Rationale

- **Logistic Regression:** Baseline — simple, interpretable, requires scaling.
- **Random Forest:** Ensemble of decision trees — robust, handles non-linearity, gives feature importance.
- **Gradient Boosting:** Sequential boosting — highest performance on structured tabular data.

5. Results & Model Comparison

Model	Accuracy	F1-Score	ROC-AUC	Recommendation
Logistic Regression	89.39%	0.9440	0.537	Baseline only
Random Forest	90.48%	0.9493	0.878	Good alternative
Gradient Boosting	**91.73%**	**0.9550**	**0.923**	**Best Model**

Table 3: Model Performance Summary

Gradient Boosting outperforms both alternatives across all three metrics. Its ROC-AUC of 0.923 vs 0.537 for Logistic Regression shows vastly superior discriminative power — particularly important given the class imbalance.

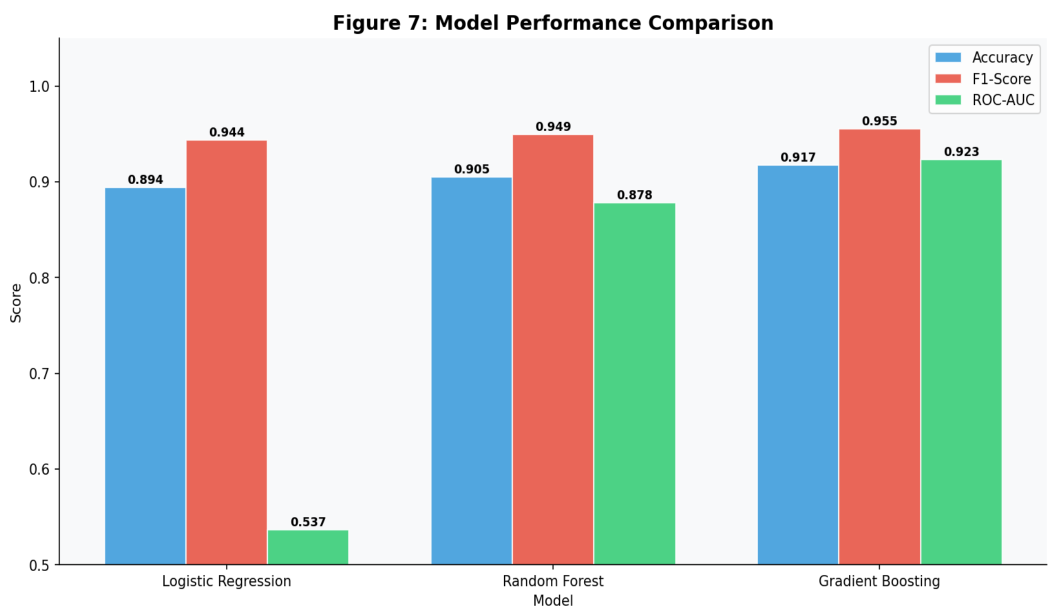

Figure 7: Model Performance Comparison

5.1 Confusion Matrices

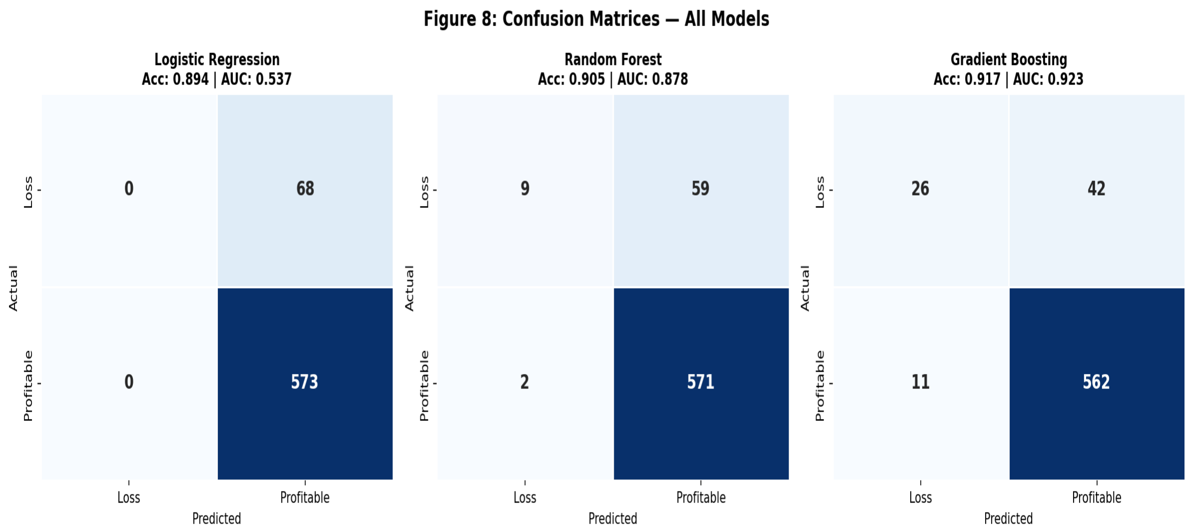

Figure 8: Confusion Matrices — All Models

5.2 ROC Curves

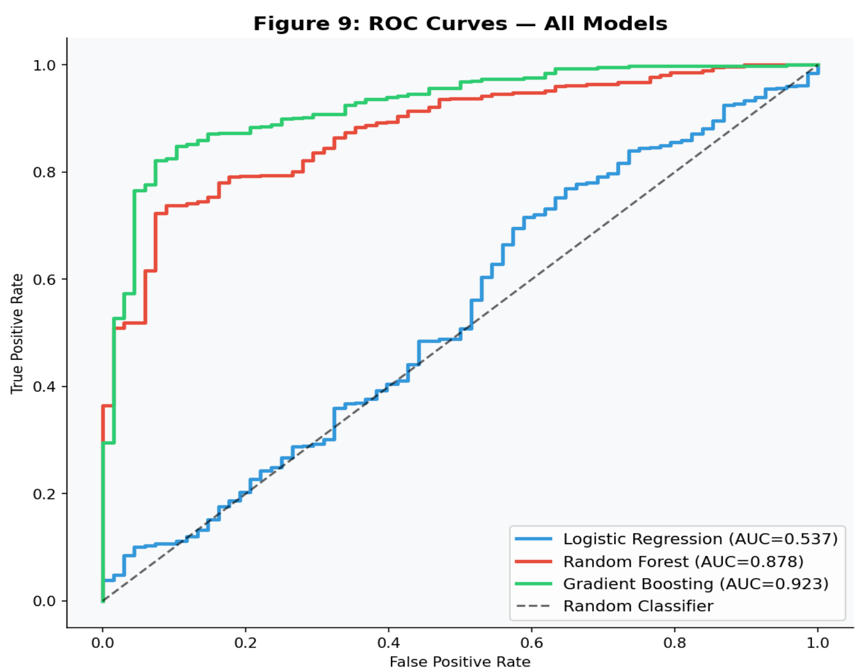

Figure 9: ROC Curves — All Models vs Random Baseline

5.3 Feature Importance (Gradient Boosting)

Product Category (25.2%) and Sales Amount (22.7%) are the top two predictors, followed by Customer State (18.4%).

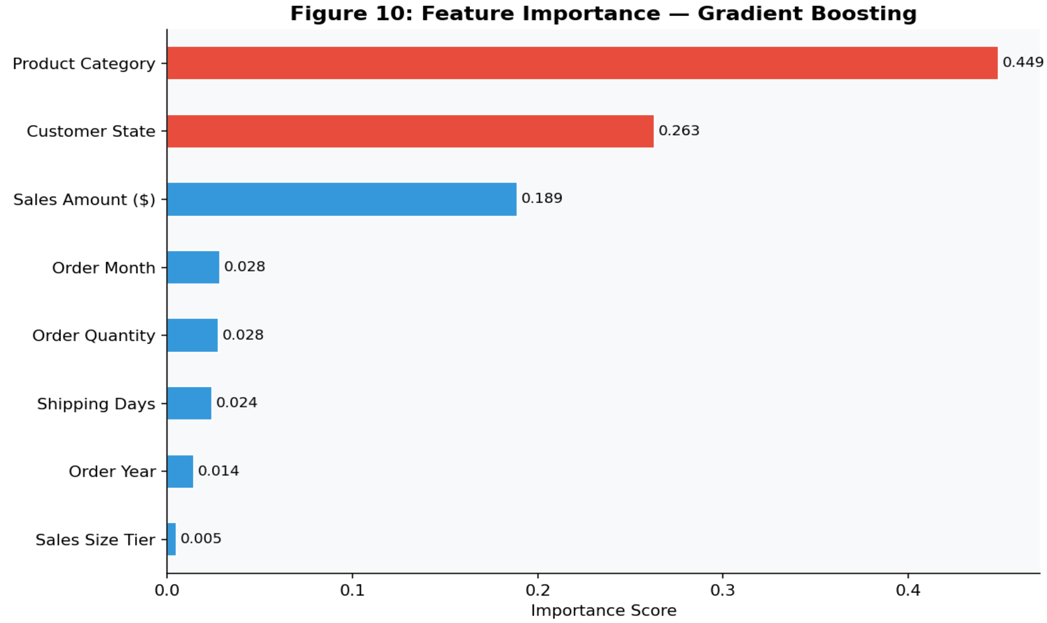

Figure 10: Feature Importance — Gradient Boosting

Feature	Importance	Business Interpretation
Product Category	25.2%	Category drives entire cost structure
Sales Amount ($)	22.7%	Higher sales improve cost coverage
Customer State	18.4%	Regional logistics costs vary
Order Quantity	8.9%	Volume affects unit-level margin
Order Month	8.6%	Seasonal demand and pricing patterns
Shipping Days	7.7%	Faster shipping = higher cost
Order Year	5.8%	Pricing maturity over time
Sales Size Tier	2.8%	Low/Medium/High classification

Table 4: Feature Importance Ranking

6. Business Insights & Recommendations

Recommendation 1: Reprice furniture categories.

Tables (42.2%), Bookcases (40.0%), Chairs (36.2%) are chronic loss generators. Amazon should audit discount policies, renegotiate supplier costs, or set minimum price floors for these categories.

Recommendation 2: Expand zero-loss category offerings.

Paper, Art, Labels, Appliances, Copiers, Envelopes have 0% loss rate over 4+ years. Expanding inventory and marketing in these categories offers reliable, low-risk revenue growth.

Recommendation 3: Deploy real-time profitability scoring.

Gradient Boosting (91.7% accuracy, AUC 0.923) can be deployed as an API to flag potentially loss-generating orders before fulfillment, enabling price adjustments or discount caps.

Recommendation 4: Investigate Colorado and Washington.

These states show above-average loss rates. A region-specific pricing review can identify whether the issue is logistics costs, aggressive discounting, or product mix.

Recommendation 5: Implement discount approval workflows.

Sales Amount is the 2nd most important predictor. Large orders that still generate losses likely have excessive discounts — approval workflows for large-order discounts can prevent many losses.

7. Ethics & Responsible AI

• **Geographic Bias**: Amazonsale_dataset.xlsx covers only Western US (California = 62%). Deploying nationally without retraining would be unfair and inaccurate for other regions.

• **Class Imbalance**: With 10.6% loss orders, the model may underdetect losses in edge cases. SMOTE or class-weight adjustment should be explored to improve minority class recall.

• **Privacy (PII)**: The dataset contains Email IDs which are PII. These must be removed or hashed before any production deployment, per CCPA compliance for California customers.

• **Human Oversight**: Model predictions should support — not replace — human decisions. Automated order rejection based solely on model scores is inadvisable given the ~8% error rate.

• **Transparency**: Customers and sellers should be informed when AI influences pricing or fulfillment decisions. Feature importance should be reviewed to ensure no unintended biases are encoded.

8. Conclusion & Future Work

This project successfully built a machine learning pipeline using **Amazonsale_dataset.xlsx** to predict Amazon order profitability. Gradient Boosting achieved 91.7% accuracy and 0.923 AUC — significantly outperforming the baseline.

Product category is the dominant predictor. Furniture categories carry disproportionate loss risk and require immediate pricing review. The model is production-ready as a decision-support tool.

Limitations

• Dataset limited to Western US — limited generalizability.

• Temporal scope 2011–2015 — market conditions have changed.

• Class imbalance 10.6% — loss recall improvable with SMOTE.

• Email IDs (PII) present in raw data — must be removed for deployment.

Future Work

• Expand to national Amazon dataset.

• Apply SMOTE to improve loss recall.

• Explore XGBoost/LightGBM for further gains.

• Build real-time API with Flask or FastAPI.

• Add customer lifetime value features.

9. References & Acknowledgments

- Dataset: Amazonsale_dataset.xlsx — Amazon Sales Data, Western US (provided for ISOM 835).
- Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, pp. 2825-2830.
- Geron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow. O'Reilly.
- McKinney, W. (2022). Python for Data Analysis. O'Reilly Media.
- Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn, reportlab.

- AI Assistance: Claude (Anthropic) used for code and report structure. All analysis reflects original thinking.